VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

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Date: 11-03-2026
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1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".

```
>_ mongosh: aviral
>_MONGOSH
> use vehicles
< switched to db vehicles
```

2. Write a MongoDB query to display all the databases.

```
> show dbs
< admin   40.00 KiB
  config  36.00 KiB
  local   40.00 KiB
vehicles>
```

3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.

```
> db.createCollection("two_wheelers", { capped: true, size: 10000, max: 100 })
< { ok: 1 }
> db.createCollection("four_wheelers")
< { ok: 1 }
vehicles>
```

4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
> db.two_wheelers.insertMany([
    {
      bike_name: "Royal Enfield Classic 350",
      model: "gear",
      category: "350cc",
      colors_available: ["black", "red", "stealth black", "signals"],
      manufacturer: "Royal Enfield",
      performance: 7,
      timestamp: new Date("2022-01-15"),
      price: 195000
    },
    {
      bike_name: "Honda Activa 6G",
      model: "gearless",
      category: "125cc",
      colors_available: ["pearl sunbeam white", "black", "blue"],
      manufacturer: "Honda",
      performance: 8,
      timestamp: new Date("2021-06-10"),
      price: 75000
    },
    {
      bike_name: "Bajaj Pulsar NS200",
      model: "gear",
      category: "200cc",
      colors_available: ["red", "black", "sport red"],
      manufacturer: "Bajaj",
      performance: 9,
      timestamp: new Date("2023-03-20"),
      price: 145000
    },
    {
      bike_name: "TVS Jupiter",
      model: "gearless",
      category: "125cc",
      colors_available: ["blue", "grey", "red", "white"],
      manufacturer: "TVS",
      performance: 7,
      timestamp: new Date("2020-08-05"),
      price: 80000
    },
    {
      bike_name: "Yamaha FZ-S V3",
      model: "gear",
      category: "150cc",
      colors_available: ["black", "blue", "sport red"],
      manufacturer: "Yamaha",
      performance: 8,
      timestamp: new Date("2022-11-01"),
      price: 125000
    }
  ])
```

```
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('69b16c85bfec7571c5f4eb58'),
      '1': ObjectId('69b16c85bfec7571c5f4eb59'),
      '2': ObjectId('69b16c85bfec7571c5f4eb5a'),
      '3': ObjectId('69b16c85bfec7571c5f4eb5b'),
      '4': ObjectId('69b16c85bfec7571c5f4eb5c')
    }
  }
```

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
db.four_wheelers.insertMany([
  {
    vehicle_name: "Maruti Suzuki Swift",
    model: "own",
    category: "car",
    variants: ["vxi", "zxi", "petrol", "diesel"],
    manufacturer: "Maruti Suzuki",
    performance: 8,
    timestamp: new Date("2022-05-15"),
    price: 800000
  },
  {
    vehicle_name: "Tata Ace",
    model: "commercial",
    category: "mini truck",
    variants: ["diesel", "cng"],
    manufacturer: "Tata Motors",
    performance: 7,
    timestamp: new Date("2021-07-10"),
    price: 650000
  },
  {
    vehicle_name: "Ashok Leyland Viking",
    model: "commercial",
    category: "bus",
    variants: ["diesel"],
    manufacturer: "Ashok Leyland",
    performance: 6,
    timestamp: new Date("2020-03-25"),
    price: 3500000
  },
  {
    vehicle_name: "Hyundai Creta",
    model: "own",
    category: "car",
    variants: ["petrol", "diesel", "vxi", "zxi", "sx"],
    manufacturer: "Hyundai",
    performance: 9,
    timestamp: new Date("2023-01-18"),
    price: 1500000
  },
  {
    vehicle_name: "BharatBenz 2823",
    model: "commercial",
    category: "heavy truck",
    variants: ["diesel"],
    manufacturer: "BharatBenz",
    performance: 7,
    timestamp: new Date("2021-09-30"),
    price: 5000000
  }
])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b16d55bfec7571c5f4eb5d'),
    '1': ObjectId('69b16d55bfec7571c5f4eb5e'),
    '2': ObjectId('69b16d55bfec7571c5f4eb5f'),
    '3': ObjectId('69b16d55bfec7571c5f4eb60'),
    '4': ObjectId('69b16d55bfec7571c5f4eb61')
  }
}
```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

```
// All two-wheelers
db.two_wheelers.find()
// All four-wheelers
db.four_wheelers.find()
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

```
> db.two_wheelers.find({}, { bike_name: 1, price: 1, _id: 0 })
< {
    bike_name: 'Royal Enfield Classic 350',
    price: 195000
  }
  {
    bike_name: 'Honda Activa 6G',
    price: 75000
  }
  {
    bike_name: 'Bajaj Pulsar NS200',
    price: 145000
  }
  {
    bike_name: 'TVS Jupiter',
    price: 80000
  }
  {
    bike_name: 'Yamaha FZ-S V3',
    price: 125000
  }
```

8. Write a MongoDB query to display two_wheelers from a particular company

```
> db.two_wheelers.find({ manufacturer: "Honda" })

< {
    _id: ObjectId('69b16c85bfec7571c5f4eb59'),
    bike_name: 'Honda Activa 6G',
    model: 'gearless',
    category: '125cc',
    colors_available: [
      'pearl sunbeam white',
      'black',
      'blue'
    ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: 2021-06-10T00:00:00.000Z,
    price: 75000
  }
```

9. Write a MongoDB query to display four_wheelers available in diesel variants

```
db.four_wheelers.find({ variants: "diesel" })
```

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

```
// Two-wheelers with performance > 5
db.two_wheelers.find( { performance: { $gt: 5 } }, { bike_name: 1, category: 1, manufacturer: 1, _id: 0 } )
// Four-wheelers with performance > 5
db.four_wheelers.find( { performance: { $gt: 5 } }, { vehicle_name: 1, category: 1, manufacturer: 1, _id: 0 } )
```

2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.

```
>_ mongosh: aviral
>_MONGOSH
> use animal
< switched to db animal
animal>
```

2. Write a MongoDB query to display all the databases.

```
> show dbs
< admin     40.00 KiB
  config    96.00 KiB
  local     40.00 KiB
  vehicles  80.00 KiB
animal>
```

3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.

```
> db.createCollection("wild_animals", { capped: true, size: 10000, max: 100 })
< { ok: 1 }
> db.createCollection("domestic_animals")
< { ok: 1 }
animal>
```

4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
db.wild_animals.insertMany([
 {
  animal_name: "Bengal Tiger",
  nature: "harm",
  favorite_foods: ["deer", "wild boar", "buffalo"],
  care_taker_name: "Rajan Kumar",
  life_span: 15,
  timestamp: new Date("2021-03-10"),
  expenses: 45000
 },
 {
  animal_name: "African Lion",
  nature: "harm",
  favorite_foods: ["zebra", "deer", "wildebeest"],
  care_taker_name: "Suresh Mehta",
  life_span: 14,
  timestamp: new Date("2020-07-22"),
  expenses: 50000
 },
 {
  animal_name: "Indian Elephant",
  nature: "harmless",
  favorite_foods: ["sugarcane", "bamboo", "bananas", "grass"],
  care_taker_name: "Rajan Kumar",
  life_span: 60,
  timestamp: new Date("2019-11-05"),
  expenses: 70000
 },
```

```
  {
    animal_name: "Red Fox",
    nature: "harmless",
    favorite_foods: ["rabbits", "birds", "berries", "insects"],
    care_taker_name: "Priya Nair",
    life_span: 5,
    timestamp: new Date("2022-01-18"),
    expenses: 12000
  },
  {
    animal_name: "King Cobra",
    nature: "harm",
    favorite_foods: ["rats", "frogs", "lizards"],
    care_taker_name: "Priya Nair",
    life_span: 20,
    timestamp: new Date("2023-06-30"),
    expenses: 8000
  }
])
```

Output -

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b1730dbfec7571c5f4eb62'),
    '1': ObjectId('69b1730dbfec7571c5f4eb63'),
    '2': ObjectId('69b1730dbfec7571c5f4eb64'),
    '3': ObjectId('69b1730dbfec7571c5f4eb65'),
    '4': ObjectId('69b1730dbfec7571c5f4eb66')
  }
}
```

5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
db.domestic_animals.insertMany([
 {
  animal_name: "Golden Retriever",
  gender: "male",
  favorite_foods: ["meat", "dog biscuits", "chicken"],
  animal_petname: "Bruno",
  life_span: 12,
  timestamp: new Date("2021-05-14"),
  expenses: 5000
 },
 {
  animal_name: "Persian Cat",
  gender: "female",
  favorite_foods: ["fish", "milk", "cat food"],
  animal_petname: "Snowball",
  life_span: 15,
  timestamp: new Date("2022-02-20"),
  expenses: 3500
 },
 {
  animal_name: "Holland Lop Rabbit",
  gender: "female",
  favorite_foods: ["carrots", "lettuce", "hay", "pellets"],
  animal_petname: "Fluffy",
  life_span: 7,
  timestamp: new Date("2023-03-11"),
  expenses: 1500
 },
```

```
  {
    animal_name: "Indian Parrot",
    gender: "male",
    favorite_foods: ["seeds", "fruits", "nuts", "corn"],
    animal_petname: "Mitthu",
    life_span: 25,
    timestamp: new Date("2020-09-08"),
    expenses: 2000
  },
  {
    animal_name: "Domestic Pig",
    gender: "male",
    favorite_foods: ["vegetables", "grains", "fruits"],
    animal_petname: "Pinky",
    life_span: 8,
    timestamp: new Date("2021-12-01"),
    expenses: 4000
  }
])
```

Output -

```
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('69b17335bfec7571c5f4eb67'),
      '1': ObjectId('69b17335bfec7571c5f4eb68'),
      '2': ObjectId('69b17335bfec7571c5f4eb69'),
      '3': ObjectId('69b17335bfec7571c5f4eb6a'),
      '4': ObjectId('69b17335bfec7571c5f4eb6b')
    }
  }
```

6. Write a MongoDB query to display all documents available in wild_animals and domastic_animals.

```
// All wild animals
db.wild_animals.find()
// All domestic animals
db.domestic_animals.find()
```

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

```
> db.wild_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })
< {
    animal_name: 'Bengal Tiger',
    expenses: 45000
  }
  {
    animal_name: 'African Lion',
    expenses: 50000
  }
  {
    animal_name: 'Indian Elephant',
    expenses: 70000
  }
  {
    animal_name: 'Red Fox',
    expenses: 12000
  }
  {
    animal_name: 'King Cobra',
    expenses: 8000
  }
```

```
> db.domestic_animals.find({}, { animal_name: 1, expenses: 1, _id: 0 })
< {
    animal_name: 'Golden Retriever',
    expenses: 5000
  }
  {
    animal_name: 'Persian Cat',
    expenses: 3500
  }
  {
    animal_name: 'Holland Lop Rabbit',
    expenses: 1500
  }
  {
    animal_name: 'Indian Parrot',
    expenses: 2000
  }
  {
    animal_name: 'Domestic Pig',
    expenses: 4000
  }
```

8. Write a MongoDB query to display domestic_animals whose life is a particular year

```
> db.domestic_animals.find({ life_span: 15 })

< {
    _id: ObjectId('69b17335bfec7571c5f4eb68'),
    animal_name: 'Persian Cat',
    gender: 'female',
    favorite_foods: [
      'fish',
      'milk',
      'cat food'
    ],
    animal_petname: 'Snowball',
    life_span: 15,
    timestamp: 2022-02-20T00:00:00.000Z,
    expenses: 3500
  }
```

9. Write a MongoDB query to display wild_animals available under a particular care_taker

```
> db.wild_animals.find({ care_taker_name: "Rajan Kumar" })

< {
    _id: ObjectId('69b1730dbfec7571c5f4eb62'),
    animal_name: 'Bengal Tiger',
    nature: 'harm',
    favorite_foods: [
      'deer',
      'wild boar',
      'buffalo'
    ],
    care_taker_name: 'Rajan Kumar',
    life_span: 15,
    timestamp: 2021-03-10T00:00:00.000Z,
    expenses: 45000
  }
  {
    _id: ObjectId('69b1730dbfec7571c5f4eb64'),
    animal_name: 'Indian Elephant',
    nature: 'harmless',
    favorite_foods: [
      'sugarcane',
      'bamboo',
      'bananas',
      'grass'
    ],
    care_taker_name: 'Rajan Kumar',
    life_span: 60,
    timestamp: 2019-11-05T00:00:00.000Z,
    expenses: 70000
  }
```

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```
// Wild animals with lifespan > 5
db.wild_animals.find(
{ life_span: { $gt: 5 } },
 { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 }
)
// Domestic animals with lifespan > 5
db.domestic_animals.find(
{ life_span: { $gt: 5 } },
 { animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0 }
 )
```

```
< {
    animal_name: 'Bengal Tiger',
    favorite_foods: [
      'deer',
      'wild boar',
      'buffalo'
    ],
    expenses: 45000
  }
  {
    animal_name: 'African Lion',
    favorite_foods: [
      'zebra',
      'deer',
      'wildebeest'
    ],
    expenses: 50000
  }
  {
    animal_name: 'Indian Elephant',
    favorite_foods: [
      'sugarcane',
      'bamboo',
      'bananas',
      'grass'
    ],
    expenses: 70000
  }
  {
    animal_name: 'King Cobra',
    favorite_foods: [
      'rats',
      'frogs',
      'lizards'
    ],
    expenses: 8000
  }
```

Output for 1st Command

```
< {
    animal_name: 'Golden Retriever',
    favorite_foods: [
      'meat',
      'dog biscuits',
      'chicken'
    ],
    expenses: 5000
  }
  {
    animal_name: 'Persian Cat',
    favorite_foods: [
      'fish',
      'milk',
      'cat food'
    ],
    expenses: 3500
  }
  {
    animal_name: 'Holland Lop Rabbit',
    favorite_foods: [
      'carrots',
      'lettuce',
      'hay',
      'pellets'
    ],
    expenses: 1500
  }
  {
    animal_name: 'Indian Parrot',
    favorite_foods: [
      'seeds',
      'fruits',
      'nuts',
      'corn'
    ],
    expenses: 2000
  }
```

Output for 2nd Command